

Physics-Informed ML and Domain Adaptation for Dark Matter Classification

Pallab Mondal | 946804 | AI Models for Physics 2025/2026

MSc Examination Project - Oral Presentation (10 slides)

Phase 1: ResNet-18 vs LensPINN vs HEALSwIn on DeepLense Model I

Phase 2: ADDA domain adaptation + real SDSS lens inference

The Astrophysical Problem

Strong gravitational lensing

Background light is bent by a foreground mass, forming Einstein rings and arcs.

Competing dark matter models

- CDM: point-mass subhalos
- Axion/Fuzzy DM: interference wiggles in arcs

Objective

Classify substructure from lensing pixels: simulated training, observational adaptation.

Architectures Compared

1. ResNet-18 - standard CNN on raw pixels (baseline)
2. LensPINN - SIS lens inversion + residual map + CNN decoder
3. HEALSwIn - HEALPix projection + Swin Transformer + physics head
4. ADDA - adversarial alignment sim -> telescope domain (Phase 2)

LensPINN: Physics-Informed Inversion

Lens equation: $\beta = \theta - \alpha(\theta, \theta_E)$

SIS deflection: $\alpha(\theta) = \theta_E * \theta / |\theta|$

Pipeline:

1. Predict θ_E from image
2. Reconstruct unlensed source I_{source}
3. Residual: $I_{\text{residual}} = I_{\text{orig}} - I_{\text{source}}$
4. Classify from [original, source, residual]

Phase 1 Results (Model I Test Set)

30,000 train / 7,500 test | 64x64 | 3 classes

Accuracy: ResNet 43.41% | LensPINN 52.35% | HEALSwIn 33.33%

Macro F1: 0.355 | 0.511 | 0.167

CDM recall: 2% | 28% | 0%

LensPINN +9.0% accuracy; residual subtraction exposes subhalo shears.

Why Physics Helps / HEALSwin Lesson

ResNet: bright Einstein ring masks tiny CDM shears -> 2% CDM recall.

LensPINN: physical background subtraction -> 28% CDM recall.

HEALSwin at 5 epochs: 33.3% mode collapse.

Physics-informed models must learn θ_E first (LensPINN flat until epoch 16).

Transformers need 30-50 epochs for stable convergence.

Phase 2: Domain Adaptation (ADDA)

Telescope images have noise, PSF blur, atmospheric effects -> domain shift.

ADDA: freeze source encoder M_s ; train target encoder M_t vs discriminator D .

M_t learns to map real images into simulation feature space.

Live demo: `python my_work/train_adda.py --limit_batches 50`

Real SDSS Targets + Inference

Downloaded via SDSS SkyServer coordinates:

- Cosmic Horseshoe (J1011+0143)
- Cheshire Cat (J1038+4849)
- Five-Image Quasar (J1004+4112)

Preprocess: grayscale, center-crop, 64x64, normalize [0,1].

LensPINN predictions: no_substructure at >99.9% (scientifically correct).

Code Architecture

my_work/models/lens_pinn.py - physics encoder + classifier

my_work/utils/physics.py - differentiable SIS inversion

my_work/train.py - supervised training

my_work/evaluate.py - metrics + confusion matrices

my_work/train_adda.py - ADDA training

my_work/classify_real_lenses.py - SDSS inference

Results: results/checkpoints/, results/plots/

Conclusions

1. LensPINN outperforms ResNet on subhalo detection via ray-tracing residuals.
2. ADDA + SDSS pipeline demonstrates sim-to-real inference path.
3. HEALSwIn shows physics-parameter learning needs longer training.

Repository: <https://github.com/Pallab9999/DeepLenseProject>

Thank you - questions welcome.